

The Singulion and the Sequence: A Methodological Review of Layfield and Davis (2026a, 2026b)

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ABSTRACT

In two recent companion papers, Layfield and Davis apply latent semantic analysis (LSA) to the Voynich Manuscript and derive two hypotheses: that the manuscript was conceived as a sequence of single-bifolium quires (“singulions”) later bound out of order, and that the intended sequence of bifolia can be recovered by maximizing textual similarity across page transitions. This review examines both hypotheses against the papers’ own published data. The empirical core of the first hypothesis, an elevated textual similarity within physical bifolia, is real and independently confirmed. The reading-order interpretation built upon it, however, is underdetermined by the data and contradicted by the authors’ own explanation of the herbal section. The second hypothesis rests on the premise that the intended order of a meaningful text maximizes lexical similarity between adjacent pages. This premise is refuted by the authors’ own calibration manuscripts, and the resequencing experiment lacks two controls that would make its reported improvements interpretable: a permutation null distribution and a calibration on deliberately shuffled known text. A similarity-based resequencing of the manuscript is nevertheless methodologically sound under one specific model of the text, namely that page-to-page similarity reflects proximity of production rather than continuity of meaning. Three concrete experiments, each executable with the authors’ existing pipeline, would discriminate between the competing interpretations.

KEYWORDS

Voynich Manuscript; latent semantic analysis; codicology; singulion; bifolium; seriation

1. Introduction

Layfield and Davis have published two closely connected studies of the Voynich Manuscript. The first (Layfield and Davis, 2026a) applies latent semantic analysis to the manuscript text and reports that the manuscript “seems to follow a coherence pattern similar to known texts.” The second (Layfield and Davis, 2026b) builds on the same computational apparatus (the semantic spaces of the two papers are, in the authors’ words, “virtually identical”) and advances two hypotheses: first, that the manuscript was not originally conceived as nested quires at all, but as a stack of single-bifolium quires (“singulions”), each intended to be read as four consecutive pages, and later bound out of order by someone who, in the authors’ formulation, was “not able to ‘read’” the text; second, that the originally intended sequence of bifolia

within a section can be recovered computationally, by selecting the permutation of bifolia that maximizes the average similarity score at the transitions between them.

These are two logically distinct claims, and they deserve separate evaluation, because they fail in different ways. The singulion hypothesis rests on a solid empirical finding whose interpretation is underdetermined: the data establish that the physical bifolium is a unit of textual coherence, but not that it is a unit of *reading*. The resequencing hypothesis rests on the methodological premise that the intended order of a meaningful text maximizes lexical similarity between adjacent pages, a premise with no foundation in computational linguistics, and one that the authors' own calibration data refute.

The intent of this review is constructive. The central measurement of Layfield and Davis (2026b) is sound and converges with independent results obtained by entirely different means (Timm and Schinner, submitted). What requires correction is the inferential superstructure, and the correction can be tested: Section 5 specifies three experiments, each executable with the authors' existing pipeline and data, whose outcomes would discriminate between the competing interpretations.

Throughout, quotations from Layfield and Davis (2026b) are identified by the paragraph numbers of the online edition, and quotations from Layfield and Davis (2026a) by the paragraph numbers of the DHQ edition.

2. What the data establish

The first research question of Layfield and Davis (2026b) is well designed in structure. The authors define three relationships between pages: confoliate (the two sides of one leaf), facing (two pages visible in an opening of the bound book), and conjoint (the two interior pages of a bifolium, which in a nested quire are separated by intervening leaves but which lie side by side on the physical sheet). They compute average LSA similarity scores for each relationship, and they calibrate the expectations against two correctly bound medieval manuscripts, the Aberdeen Bestiary and a copy of the *Speculum Humanae Salvationis*. In the calibration texts, confoliate and facing scores align while conjoint scores fall below them. This is the signature of a text written continuously through nested quires. In the Voynich Manuscript the pattern inverts: conjoint (0.433) and confoliate (0.383) scores are statistically indistinguishable ($p = 0.168$), while facing scores (0.298) fall significantly below both ($p = 0.005$; their Tables 10–11).

The finding itself is robust. It appears not only at the manuscript-wide level but per quire (their Table 8), including throughout the herbal section, and at a finer decomposition (their Table 12), where the conjoint average (0.431) exceeds both confoliate averages. It also converges with results obtained by an entirely different method: substring-frequency markers on Currier-diagnostic vocabulary, that is, on the high-frequency word forms which the LSA pipeline removes as stop words, detect the same within-bifolium coherence with strong per-corpus statistical significance (Timm and Schinner, submitted). Two pipelines with opposite preprocessing, one operating on the vocabulary band the other discards, measure the same asymmetry. The observation that pages sharing a physical sheet are textually more similar than pages that merely face each other in the current binding can therefore be regarded as established.

Everything at issue in this review concerns what follows from that observation.

3. The singulion hypothesis: a real pattern, an underdetermined inference

From the inverted score pattern, Layfield and Davis (2026b) infer that each bifolium was “meant to be read” as four consecutive pages, that the manuscript was conceived as a sequential stack of singulions, and that the current nested binding is a fifteenth-century error by a binder who could not read the text. The inference faces four problems, each arising within the paper’s own data and text.

3.1. *The missing absolute scale*

LSA cosine similarities in a TF-IDF weighted, dimension-reduced space have no natural chance level, and the paper computes no random-pair baseline at any point. The analysis therefore establishes an *ordering* of the three relationship categories (facing below confoliate and conjoint), but does not locate any of them on an absolute scale. This matters because the misbinding thesis makes a specific absolute prediction: if the current configuration was produced by a binder shuffling a stack of singulions, then currently-facing pages are essentially random neighbors and should score *at* chance. If instead facing pages score above chance but below conjoint pages, the scrambling account requires qualification. The paper’s data cannot distinguish these cases.

The missing measurement has since been supplied by the marker-based methodology: applying an identical rank statistic with an exact null distribution to bifolium partners and to binding-adjacent cross-bifolium pairs shows partners far below the random expectation and binding-adjacent pairs at or slightly above it (Timm and Schinner, submitted). The result is consistent with the authors’ relative finding and completes it: whatever criterion governed the nesting of bifolia into quires, it was uncorrelated with textual similarity. Both the misbinding account and the production-order account survive this datum; the point is that the paper’s own analysis could not have detected the difference, and the conclusion is stated as if it had.

3.2. *The herbal section contradicts the paper’s own explanation*

For the herbal section, the authors explain the low facing scores by topical independence: “each page is a distinct botanical topic (as also inferred from the illustrations, which change with each page)” (para. 35). This claim is attributed to the first paper (Layfield and Davis, 2026a), but the first paper does not in fact make this claim about the herbal section. Layfield and Davis conclude that “each page is likely a different topic” for the *Cosmological and Astronomical* sections (their Section 5, point 2), not for the herbal section. For the herbal section, the first paper attributes the low scores to the Currier A/B transitions, noting that “77%” of the lowest scores “were Herbal → Herbal page comparisons, virtually all of which were near or on a page break” (their Section 4, para. 34). The “distinct botanical topic” interpretation, applied to the herbal section only in the second paper, provides the justification for excluding the herbal pages from resequencing—but it rests on a misreading of the first paper’s own findings.¹

Even granting the per-page-topic interpretation for the sake of argument, it proves too much. Under the singulion reading, the two conjoint pages of a bifolium are sim-

¹The first paper’s herbal discussion centres on A/B transitions and low discourse-segmentation scores, not on per-page topical independence. The per-page-topic interpretation appears for the first time in the second paper’s para. 35.

ply consecutive pages, and consecutive herbal pages treat equally distinct plants. If topical independence explains why facing pages are dissimilar, it equally predicts that conjoint pages should be dissimilar. Yet the paper’s own Table 8 reports the conjoint/facing differential throughout the herbal quires. Even in Quires 1 and 3, which contain exclusively Currier A bifolia and therefore cannot be affected by A/B mixing, the differential is clear (Quire 1: conjoint 0.399 against facing 0.246; Quire 3: conjoint 0.394 against facing 0.268). The topical explanation is thus invoked to account for one set of scores and silently suspended for the other set on the same pages.

The differential demands a mechanism that does not operate through topic. One such mechanism exists in the literature, and it is specific rather than ad hoc: if new text was generated by copying and modifying words already written nearby, as the self-citation process describes (Timm, 2014; Timm and Schinner, 2020), then the two pages of a physical sheet, written in the same session on the same working surface, share generation history regardless of what their illustrations depict, while pages on different sheets do not. Same-sheet production predicts exactly the conjunction the paper reports and cannot explain: elevated conjoint similarity *and* depressed facing similarity, in a section whose pages are topically independent. The reading-order interpretation, by contrast, predicts the elevation only through topical or discursive continuity, and topical continuity is what the authors themselves deny the herbal section.

The mechanism carries a documented refinement that will matter below. When starting a new page, the scribe apparently preferred the last completed sheet as the source (Timm, 2014, p. 16; restated in Timm and Schinner, 2020, p. 11): for glyph groups occurring seven or eight times in the manuscript, co-occurrence on subsequent sheets (37 cases) far exceeds co-occurrence on the two sides of a single leaf (5 cases). Further examples—rare words bridging consecutive sheets, and parallel word forms occupying matching positions on consecutive zodiac folios across different bifolia and quires—are documented in Timm (2014), pp. 8, 16. Such cases refine, rather than contradict, the bifolium-as-production-unit account: the bulk of the similarity is within-bifolium, but a localized bridging signal arises at sheet starts, where the new working surface lacks an immediate source. This is also consistent with the binding-adjacent pairs measured at or slightly above the random expectation (Section 3.1), wherever the binding happens to preserve production adjacency.

3.3. *The alternatives considered omit the standing alternative*

The paper weighs the singulion interpretation against two alternatives: chance, and limited vocabulary (para. 56). Both are reasonably dismissed on the grounds that neither explains the low facing scores. But the list omits the one alternative that explains both halves of the pattern, the elevated conjoint scores as well as the depressed facing scores, without positing a misreading binder: bifolium-wise production of the text itself. This alternative was not unknown to the authors. The bifolium-as-functional-unit question, together with marker evidence for bifolium-level textual consistency, was discussed publicly in December 2022, in a forum thread prompted by one co-author’s own conference remarks on bifolium-wise scribal production (Feaster, 2022), and synthesized shortly after under the label ‘BAAFU’ (bifolio as a functional unit) by Pelling (2022). And their own first paper generated a control text with the published self-citation software and assessed its behavior in these words: the result “might have been viewed as evidence in favour of concluding that the manuscript has no semantic content; however, this is offset somewhat by our findings regarding transitions from

Language A to Language B” (Layfield and Davis, 2026a, para. 46). A hypothesis that a paper’s own control experiment instantiated, and that its companion paper concedes might have been viewed as supported, belongs in the alternatives paragraph of the successor paper. Its absence is complete: neither the control, nor the production-process explanation, nor the underlying literature appears anywhere in Layfield and Davis (2026b), and the singulion interpretation is therefore left standing unopposed against two obviously weaker alternatives.

3.4. *The validation is borrowed from a paper this paper contradicts*

Paragraph 30 of Layfield and Davis (2026b) invokes the earlier study as validation: the demonstration that the manuscript “behaves in a very similar way to known texts . . . in the coherence and flow of individual sections.” But that demonstration was performed on the manuscript *in its current binding order*, that is, in the order the present paper declares to be a scrambling of the intended sequence. The two results cannot both carry their assigned weight. If the current order exhibits the coherence and flow of a correctly bound text, it is not detectably scrambled; if the depressed facing scores of the present paper reveal the scrambling, then the earlier coherence finding requires reinterpretation. The one available reconciliation is that the earlier coherence measurements were dominated by within-page and within-sheet adjacencies, with the corrupted bifolium-boundary transitions diluted in the averages. That reconciliation is plausible, but it concedes that the “coherence and flow” was local repetition tied to the physical sheet rather than topical continuity across a section, and it thereby transfers the first paper’s central result to the production account.

3.5. *Summary of the singulion hypothesis*

None of this diminishes the finding of Section 2. The data establish that the bifolium is a unit of textual coherence. Two hypotheses are consistent with that: the bifolium as intended *reading* unit, and the bifolium as *production* unit. Every observation in the paper that discriminates between them favors the second: the differential in a topically independent section (Section 3.2), the behavior of binding-adjacent pages at the random baseline (Section 3.1), and the fate of the borrowed validation (Section 3.4).

A further observation, not discussed in the paper, points in the same direction. The large foldout sheets—most notably the six-panel Rosettes sheet (Quire 14, ff. 85–86)—contain text panels written in opposing orientations on the reverse side: three panels are upright, while the remaining three must be rotated 180° to be read. This is consistent with a scribe who planned the eventual folding and binding structure from the outset, writing each panel in the orientation required by its position in the bound codex. A loose singulion intended to be read as a standalone sheet would have all text in one orientation.

4. The resequencing hypothesis: a method without a foundation

The second research question of Layfield and Davis (2026b) asks whether the intended sequence of bifolia can be recovered by exhaustively permuting the bifolia of a quire and selecting the permutation that maximizes the average facing score at the transitions. Applied to Quire 13 (120 permutations) and Quire 20 (720 permutations), the

procedure yields orderings that improve on the current structure by 12.1% and 20% respectively, which the paper presents as candidate reconstructions of the intended reading order.

4.1. *The premise*

The procedure presupposes that the intended order of a meaningful text maximizes lexical similarity between adjacent pages. No such principle exists in computational linguistics. What makes page $n+1$ follow page n in a natural-language text is discourse structure (anaphora, connectives, entity continuity, narrative progression), not vocabulary overlap. Topics recur non-monotonically: a text returns to a theme, alternates subjects, interleaves digressions. Lexical similarity between pages is consequently a *grouping* signal (which pages share subject matter), not a *sequencing* signal (which page comes next), and the closest ordering task in natural language processing, sentence and paragraph ordering, is attacked with discourse- and entity-based features for the simple reason that bag-of-words similarity performs poorly at it. Medieval book producers presupposed the same fact in practice: catchwords and quire signatures exist because the correct order of sheets is *not* recoverable from content, even for a perfectly readable Latin text.

4.2. *The authors' own calibration refutes the premise*

The paper's benchmark data demonstrate the problem directly. In the correctly bound Aberdeen Bestiary, the per-quire facing averages span 0.138 to 0.466 (their Table 6): the *true* order of Bestiary Quire 3 produces lower adjacent-page similarity than the allegedly scrambled Voynich Manuscript does manuscript-wide (0.298). A great many permutations of Bestiary Quire 3 would outscore its correct order under the paper's objective. The Speculum provides the converse demonstration. The authors highlight the distant conjoint pair 39v | 46r scoring 0.498 because both pages treat the Last Judgement; it follows that a similarity-maximizing resequencing of the correctly ordered Speculum would move those pages together, that is, away from the true order. On the paper's own calibration manuscripts, the maximization objective neither preserves correct orders nor distinguishes them from incorrect ones.

4.3. *The missing controls were described in the authors' own bibliography*

Two controls would make the reported improvements interpretable, and both are absent. The first is the permutation null distribution: the procedure evaluates all 120, respectively 720, orderings, so the full score distribution is computed as a byproduct, yet the paper reports only the maximum. The relevant statistic is where the *current* order ranks within that distribution. Under the paper's own thesis of a binder shuffling singulions he could not read, the current order should rank near the median; under the alternative that the binding partially preserves the production sequence, it should rank high. This single number, already computed and discarded, is a cleaner test of the misbinding thesis than anything the paper reports. In its absence, the headline improvements of 12.1% and 20% carry no information: the maximum of hundreds of draws from any noisy score distribution exceeds a typical draw, so improvements of this magnitude are expected even under the null hypothesis that no intended order exists at all.

The second missing control is a calibration on deliberately shuffled known text: scramble a Bestiary quire, run the maximization, and measure how reliably, and by what margin of improvement, the true order is recovered. This design was not unknown to the authors; their own first paper describes Reddy and Knight (2011) investigating the page order of the manuscript with exactly such a control, reporting that scrambled adjacent-page similarity scores fall “close to 0” (Layfield and Davis, 2026a, para. 13). Reddy and Knight (2011) is absent from the second paper.

The paper itself signals the indeterminacy of its results: for Quire 13, the authors entertain the *second*-best permutation as possibly the correct one on codicological grounds (paras. 52–54). A procedure whose output must be adjudicated by external evidence has conceded that it does not determine the answer; the external evidence is then doing the work the computation was credited with.

4.4. *No within-manuscript resequencing can validate itself*

The choice of test sections compounds the missing-controls problem. The resequencing was applied to Quires 13 and 20, which are, by the paper’s own Table 8, the two most homogeneous, highest-scoring sections of the manuscript, where every permutation scores well and the maximization cannot conspicuously fail. The natural alternative, the herbal quires, where Currier A and Currier B bifolia interleave within single quires (a configuration the paper itself calls “an unprecedented structure”), was excluded on the ground of topical independence (para. 35), the explanation already contradicted by the paper’s own conjoint scores (Section 3.2).

Remarkably, the second paper proposes resequencing by maximizing LSA similarity scores without any discussion of how the Currier A/B distinction—whose “magnitude” their own first paper called “surprising” (Layfield and Davis, 2026a, para. 34)—affects those scores. The dominant signal in their LSA analysis is left unaddressed in the method that depends on it.

It must be said clearly, however, that the herbal experiment would not have supplied a valid test either, for the opposite reason: it, too, could not fail. The differences between Currier A and Currier B have been defined, since Currier (1976), precisely by glyph- and word-frequency profiles (“Final [*dy*] is very high in Language B; almost non-existent in Language A,” among further contrasts of the same kind), and these profiles dominate any lexical similarity metric. A similarity-maximizing permutation of interleaved A and B bifolia therefore deinterleaves them by construction; the outcome would present itself as a dramatic confirmation of misbinding while being predetermined by the objective function. Even the apparently external success criterion, the paleographically established scribal attributions, is contaminated, since scribe and language covary in the herbal section. The general lesson is that no within-manuscript resequencing experiment can validate itself: the internal criteria are either uninformative (homogeneous sections) or built into the objective (heterogeneous ones). Validation must come from outside the manuscript. This is why the shuffled known-text calibration of Section 4.3 should be regarded as the missing foundation of the paper’s method rather than as a desirable supplement to it.

4.5. *The precedent, and the interpretation under which the method is sound*

Recovering the original arrangement of the manuscript from textual similarity is not a new idea. Timm and Schinner (2020) ordered the sections by the frequency of the token *<chedy>* and argued that the current arrangement deviates from the original one; the authors’ own first paper summarizes this accurately in a footnote: the alternative ordering “would place Herbal A and Herbal B apart from each other, with several sections in between” (Layfield and Davis, 2026a, note 12). Zandbergen (2003) computed page-to-page correlations from shared vocabulary more than two decades ago, and misbinding proposals for specific quires have long circulated in the research community (Pelling, 2006). Against this record, the second paper’s conclusion claims novelty for the misbinding thesis itself: “The novel idea expressed in this paper that 1) the entire manuscript is bound out of order...” (para. 58). What is new is the singulion format and the reading-order interpretation, and the distinction should have been drawn.

The precedent matters beyond priority, because it identifies the one model of the text under which similarity-maximizing resequencing is methodologically sound. Ordering objects by pairwise similarity is *seriation*, and seriation is valid when the data carry a monotone one-dimensional gradient, as in stratigraphy or stylistic drift. Under the self-citation model, the manuscript’s vocabulary constitutes exactly such a gradient: each sheet is copied, with modification, from its recent predecessors, a sheet-to-sheet linkage documented at the token level a decade before the LSA studies (Section 3.2), so textual similarity decays monotonically with production distance, and the permutation maximizing transition similarity approximates the *production* sequence. Under the reading-order model, no such monotonicity exists, as the authors’ own Bestiary and Speculum data confirm (Section 4.2). The resequencing experiment of Layfield and Davis (2026b) is therefore coherent only under the one interpretation of the similarity scores that the paper does not consider, and unfounded under the interpretation it adopts.

5. Three discriminating experiments

Each of the following can be executed with the authors’ existing pipeline and data, and each yields a different outcome under the reading-order and production-order interpretations.

Experiment 1: the random-pair baseline. Compute the LSA similarity distribution for randomly drawn same-section page pairs, and locate the facing, confoliate, and conjoint averages on it. Reading-order misbinding predicts facing scores at the random baseline; a binding order anticorrelated with production predicts facing scores at or below it; partial preservation of production order predicts facing scores above it.

Experiment 2: the shuffled-quire calibration and the permutation null. Scramble the bifolia of a Bestiary or Speculum quire, run the exhaustive maximization, and report (a) how often and by what margin the true order is recovered, and (b) for the Voynich quires, the rank of the current binding order within the full permutation-score distribution already computed. If the current Voynich order ranks near the median while shuffled known texts are reliably recovered, the misbinding thesis gains support; if the true Bestiary order is *not* reliably recovered, the maximization objective is uninformative about reading order and the 12.1% and 20% improvements are artifacts

of taking a maximum.

Experiment 3: the held-out gradient test. For the herbal bifolia, compute pairwise similarity with the Currier-diagnostic vocabulary families excluded from the feature set (the $\langle ed \rangle / \langle dy \rangle$ finals and related forms that have defined the A/B distinction since Currier, 1976), and order the bifolia by similarity maximization on the residual vocabulary alone. Then test whether the held-out features vary monotonically along the recovered order. Here the two interpretations diverge. If Currier A and B are discrete systems (two languages, two codes, or two scribes' independent conventions), nothing constrains the residual vocabulary to carry the same axis: the recovered order should at best form two blocks with arbitrary internal arrangement, and the held-out $\langle ed \rangle$ rate should follow a step profile. If instead the two corpora are stages of one continuously evolving system (Timm and Schinner, 2020), the gradient pervades the vocabulary beyond its defining markers, and the held-out rate should vary monotonically along the recovered order, with intermediate folios between the poles. Unlike reunification of the corpora, which any similarity metric guarantees because the A/B frequency profiles dominate lexical similarity, monotonicity of an excluded feature along an order computed without it is a prediction that can fail, in either direction.

6. Conclusion

The two hypotheses of Layfield and Davis (2026b) merit asymmetric verdicts. The measurement underlying the singulion hypothesis stands: pages sharing a physical bifolium are textually more coherent than the current binding order would suggest, a result now established by two independent methodologies and robust across sections. As evidence that the bifolium was the manuscript's *unit*, the finding is secure; the further inference that it was a unit of *reading*, produced as a stack of singulions for sequential consumption and scrambled by an uncomprehending binder, is underdetermined by the data and strained by the paper's own herbal results. The resequencing hypothesis should be set aside until the maximization objective has been validated on shuffled known texts and the current order's rank in the permutation distribution has been reported; on the authors' own calibration data, the objective does not track reading order even where reading order is known.

In closing, the authors' central finding is more interesting than the interpretation they have placed on it. A manuscript whose textual coherence tracks its physical sheets rather than its binding sequence is telling us something about how it was made. The production-level reading requires neither a binder who could not read, nor a lost intended sequence, nor semantic assumptions, and it predicts the elevated conjoint scores, the depressed facing scores, the herbal differential, and the seriability of the sections from a single documented mechanism. The three experiments above would show whether that economy of explanation is also empirical adequacy. Their execution would be welcome.

Disclosure statement

No potential conflict of interest was reported by the author.

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